IS492 · SP26

# sust-AI-naible

*Multi Agent Cloud Carbon Optimization*

**Team Carbon Emissions**

**Deployed URL: https://sustainable-demo.streamlit.app**

**GitHub:** github.com/IS492-SP26/team-project-carbon-emissions

Spring 2026 | Checkpoint #3 |

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# Problem Statement & Targeted Users

## The Problem

Data centers consume ~1% of global electricity (205 TWh/year) accelerating with AI.

EU's CSRD now legally requires ~50,000 companies to disclose Scope 1/2/3 cloud emissions.

Existing tools (AWS, Google Cloud Carbon Footprint) only measure but none close the loop by acting and verifying savings.

## The Gap

No tool today both executes optimizations AND verifies savings but sust-AI-naible does both.

## Success = Verified Savings

- ✓ 10–40% carbon reduction via time/region shifting
- ✓ Counterfactual MRV with confidence intervals
- ✓ Traceable evidence chain for every claim

Targeted Users: Cloud engineers, sustainability officers, enterprises under CSRD compliance

# Live Demo & Walkthrough

**SENSE**
Ingestor generates synthetic workloads

**MODEL**
Carbon Accountant calculates $kgCO_2e$

**DECIDE**
Planner → Governance negotiation

**ACT**
Executor applies changes + tickets

**VERIFY**
Verifier: counterfactual MRV

## Dashboard Highlights

📊 **Carbon Analysis** Emissions by region, team & hour with live Plotly charts

🤖 **The Debate** Chat style Planner↔Governance negotiation transcript

🔍 **Evidence Explorer** Click any verification to inspect full counterfactual chain

💬 **Ask the Agent** Interactive Q&A about your pipeline results

🌍 **The Impact** Savings as real world equivalencies + business value

⚠️ **Error Handling** Mock LLM fallback; deterministic numbers

# Design Decisions & Challenges

## AI Models Used

**GPT-4o-mini (optional):** Rationale generation, Planner↔Governance negotiation, developer nudges. Chosen for cost-efficiency on text-only tasks.

**Mock LLM (built-in):** Contextually appropriate responses, system works fully offline.

## Key Design Choices

**AI + Determinism boundary:** LLM explains; never calculates numbers.

**Multi-objective optimization:** Balances carbon + cost + latency (carbon-only causes 2–3× cost spikes).

**Governance gating:** Every batch approved before execution.

## User Data Collected

Synthetic workload logs, carbon intensity time series, optimization outcomes, agent reasoning traces, and gamification leaderboard data is all written to local CSV/JSON files in data.

## Challenges & Solutions

**Prompt engineering:** Compared prompts to find most effective rationale generation.

**LLM reliability:** Built mock fallback so correctness never depends on API availability.

# Safety, Privacy & Reliability

## 🔒 No PII Collected

The system operates entirely on synthetic workload data. No real user identities, credentials, or personal information are stored anywhere.

## 🔑 API Key Safety

API keys are loaded via .env and never committed to the repo. .env.example shows structure without real values. .gitignore excludes .env.

## ✅ Deterministic Verification

All carbon and cost numbers are computed deterministically which is never generated by an LLM ensuring reproducibility and auditability.

## 🌐 Offline First Reliability

Built in mock LLM ensures the full pipeline runs without network access or an OpenAI key. 50 automated tests cover all core modules.

# Next Steps — Checkpoint #4

**CP4 Goal: Collect real user data to substantiate our success claims with evidence.**

01

### Time-Saving Study

Recruit 5–10 cloud engineers. Have them estimate carbon savings manually vs. using our tool. Record time on task and accuracy. Hypothesis: ≥50% time reduction.

02

### Accuracy Validation

Compare our counterfactual MRV estimates against industry baselines (CarbonScaler, Cloud Carbon Footprint). Report RMSE and confidence interval overlap.

03

### User Satisfaction Survey

Post-demo Likert survey with cloud engineers, researchers, and professors who use cloud for academic workloads. Measure usefulness, trust in AI rationales, and dashboard clarity. Aim: ≥4/5 average on 'Would you use this for compliance reporting?'

04

### Feature Additions

Real cloud provider API integration, persistent user sessions, export to CSRD ready report format, and agent feedback loop for continuous improvement.